

PALMA DE MALLORCA AP, Spain

WMO: 083060

Lat: 39.561N

Lon: 2.737E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.7	1.9	-1.8	3.3	5.1	-0.2	3.7	5.0	12.2	13.6	10.8	13.3	0.8	40	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.8	33.7	22.3	32.1	22.5	30.9	22.5	25.5	29.7	24.7	29.1	24.1	28.6	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.1	28.0	23.2	18.0	27.6	22.7	17.4	27.2	78.1	29.7	75.0	29.2	72.3	28.5	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	8.9	7.9	-1.9	37.3	1.0	1.5	-2.6	38.4	-3.2	39.3	-3.8	40.1	-4.5	41.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-2.2	27.1	1.0	1.3	-2.9	28.0	-3.5	28.8	-4.0	29.5	-4.7	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	10.1	10.1	11.9	14.5	18.1	22.4	25.2	22.4	19.0	14.1	11.3		
		DBStd	6.06	2.46	2.54	2.41	2.19	2.42	2.54	1.97	1.91	2.35	2.49	2.84		
		HDD10.0	84	28	27	9	1	0	0	0	0	0	3	16		
		HDD18.3	1208	254	232	199	116	34	2	0	1	22	130	218		
		CDD10.0	2672	32	29	68	135	252	371	471	482	371	279	126		
		CDD18.3	753	0	0	0	1	27	123	213	223	122	43	2		
		CDH23.3	7237	0	0	2	11	186	1139	2316	2392	958	231	4		
		CDH26.7	2489	0	0	0	0	26	347	915	962	216	24	0		
(14)	Wind	WSAvg	3.0	2.8	2.9	3.2	3.4	3.2	3.3	3.2	3.1	2.9	2.7	2.9		
(15)	Precipitation	PrecAvg	610	60	48	36	49	37	15	4	25	70	82	100		
(16)		PrecMax	884	162	109	79	157	220	48	33	113	253	192	303		
(17)		PrecMin	327	0	4	4	4	3	0	0	8	11	4	0		
(18)		PrecStd	145	40	32	22	38	48	16	8	28	51	48	65		
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	19.1	19.7	22.1	25.0	29.4	34.5	36.1	35.8	32.2	29.0	23.7	19.8		
(20)		0.4% MCWB	14.0	14.0	14.6	15.9	18.3	21.9	21.3	22.5	22.8	20.9	18.4	15.4		
(21)		2% DB	17.6	18.0	20.1	22.8	26.9	32.0	33.7	33.5	30.1	27.0	22.1	18.3		
(22)		2% MCWB	13.4	13.0	13.9	15.2	17.8	21.2	22.0	22.9	22.6	21.1	17.5	14.6		
(23)		5% DB	16.4	16.8	18.9	21.4	25.2	30.1	32.0	32.0	29.0	25.8	20.9	17.5		
(24)		5% MCWB	13.0	12.7	13.6	15.0	17.3	21.1	22.3	23.1	22.1	20.5	16.8	14.3		
(25)		10% DB	15.5	15.8	17.8	20.1	24.0	28.7	30.8	30.9	27.9	24.4	19.8	16.7		
(26)		10% MCWB	12.4	12.2	13.2	14.8	17.3	20.6	22.3	23.1	21.8	19.7	16.1	13.8		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	15.3	15.1	16.4	17.9	20.8	25.3	26.1	26.7	25.1	23.4	19.6	16.6		
(28)		0.4% MCDB	17.4	18.0	19.4	21.3	24.9	29.8	30.0	30.6	29.0	26.3	21.9	18.4		
(29)		2% WB	14.4	14.2	15.3	16.8	19.7	23.6	25.0	25.8	24.2	22.2	18.5	15.6		
(30)		2% MCDB	16.6	16.6	18.2	20.4	23.8	28.2	29.5	30.0	28.2	25.3	21.0	17.5		
(31)		5% WB	13.7	13.4	14.6	16.1	19.0	22.5	24.4	25.1	23.4	21.2	17.5	15.0		
(32)		5% MCDB	15.9	15.8	17.4	19.7	23.3	27.4	28.9	29.4	27.5	24.5	20.1	17.0		
(33)		10% WB	12.8	12.6	13.9	15.5	18.3	21.7	23.7	24.4	22.6	20.4	16.5	14.1		
(34)		10% MCDB	15.1	15.1	16.7	19.0	22.5	26.6	28.4	28.9	26.7	23.7	19.2	16.3		
(35)	Mean Daily Temperature Range	MDBR	10.1	10.3	11.0	10.9	10.9	11.3	11.2	10.8	10.1	9.9	9.5	9.7		
(36)		5% DB MCDBR	11.1	11.9	12.9	13.0	12.8	13.5	13.2	12.5	11.0	11.1	10.7	10.4		
(37)		5% WB MCWBR	7.9	8.2	8.0	7.2	6.1	6.3	5.8	5.9	5.5	6.3	7.1	7.5		
(38)		5% WB MCDBR	10.2	10.7	11.1	11.3	11.0	11.9	11.0	10.8	9.9	10.2	9.8	9.7		
(39)	Clear-Sky Solar Irradiance	taub	0.339	0.362	0.407	0.429	0.450	0.459	0.475	0.466	0.446	0.415	0.375	0.344		
(40)		taud	2.482	2.407	2.267	2.227	2.206	2.220	2.187	2.219	2.262	2.329	2.389	2.461		
(41)		Ebn at Noon	827	852	844	848	835	826	808	805	795	780	768	787		
(42)		Edh at Noon	79	97	124	137	143	141	144	136	123	104	86	76		
(44)	All-Sky Solar Radiation	RadAvg	2.19	2.99	4.40	5.63	6.53	7.35	7.19	6.30	4.71	3.42	2.29	1.92		
(45)		RadStd	0.13	0.34	0.36	0.32	0.38	0.29	0.27	0.25	0.25	0.25	0.18	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+97	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page